

Question	Answer	Mark	Mark scheme	Additional guidance
21	3 : 4	P1	starts process eg $\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$ oe	Formal geometric reasoning relating to congruent and similar triangles is not required
		P1	for process to find $\overrightarrow{OM} = \mathbf{a} + \frac{1}{2} “(\mathbf{b} - \mathbf{a})”$ oe $(= \frac{1}{2}(\mathbf{a} + \mathbf{b}))$	
		P1	for process to find $\overrightarrow{AP} = -\mathbf{a} + \frac{3}{5} “(\frac{1}{2}\mathbf{a} + \frac{1}{2}\mathbf{b})”$ oe or (indep) for $\overrightarrow{AN} = -\mathbf{a} + “k”\mathbf{b}$	
		P1	process to find “ k ” using $\overrightarrow{AN} = -\mathbf{a} + “k”\mathbf{b}$ as a multiple of $\overrightarrow{AP}$	
		A1	cao	
			ALTERNATIVE	
		P1	for producing OM to C such that AC is parallel to OB	
		P1	for process to show that $MC = OM$, using congruent triangles ACM and BOM	
		P1	for process to find PC as a multiple of $OM/5$ $(= 7OM/5)$	
		P1	for process to find ON as a multiple of $AC(OB)$ $(= 3OB/7)$ using similar	
		P1	triangles ACP and NOP	
		A1	cao	